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ORIGINAL PAPER

Marine macroalga-associated heterotrophic *Bacillus velezensis*: a novel antimicrobial agent with siderophore mode of action against drug-resistant nosocomial pathogens

Kajal Chakraborty[1] · Aneetta Francis[1,2] · Rekha Devi Chakraborty[3] · Sumayya Asharaf[1] · Vinaya Kizhakkepatt Kizhakkekalam[1] · Silpa Kunnappilly Paulose[1]

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Abstract

Increased prevalence of microbial resistance and development of drug-resistant pathogens have triggered an urge among researchers to discover potential antimicrobial compounds, particularly from the marine habitat. The present study highlights the cultivable diversity and bioactivities of heterotrophic bacteria associated with marine macroalgae of southeast Indian coastal region. Culture-dependent isolation method resulted in 40 isolates, in which greater part of the isolates represented *Gammaproteobacteria* (62%) followed by that comprised of the phylum *Firmicutes*. One of the most active strains isolated from a macroalga, *Laurencia papillosa*, was characterized based on the integrated phenotypic and genotypic analysis as *Bacillus velezensis* MBTDLP1 MTCC 13048, with an inhibition zone of about 35 mm against methicillin-resistant *Staphylococcus aureus* (MRSA), was selected for bioprospecting studies. Type-I *pks* gene (MT394492) of 700 bp could be amplified from the heterotrophic *B. velezensis*. The bacterium exhibited siderophore production and possessed genes implicated in the biosynthesis of siderophore type of metabolites exhibiting 99% similarity with other GenBank sequences in BLAST search. *B. velezensis* exhibited promising anti-infective properties against methicillin-resistant *Staphylococcus aureus* (minimum inhibitory concentration 15 µg/mL), and the activities were positively correlated ($r^2 > 0.9$) with iron-chelating activities. Chemical investigation of the organic extract of *B. velezensis* MBTDLP1 characterized a macrocyclic polyketide exhibiting prospective antibacterial potential against methicillin-resistant *S. aureus* (MIC 0.38 µg/mL), than that exhibited by positive control chloramphenicol (6.25 µg/mL). Significant antibacterial activity against drug-resistant bacteria combined with the presence of genes coding for bioactive secondary metabolites revealed that this marine symbiotic bacterium could be used against emerging antibiotic resistance.

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✉ Kajal Chakraborty
kajal.chakraborty@icar.gov.in; kajal_cmfri@yahoo.com

[1] Marine Biotechnology Division, Central Marine Fisheries Research Institute, P.B. No. 1603, Ernakulam North, Cochin, India

[2] Department of Plant Biotechnology, College of Agriculture, Kerala Agricultural University, Vellayani, Thiruvanthapuram 695522, India

[3] Crustacean Fisheries Division, Central Marine Fisheries Research Institute, P.B. No. 1603, Ernakulam North, Cochin, India

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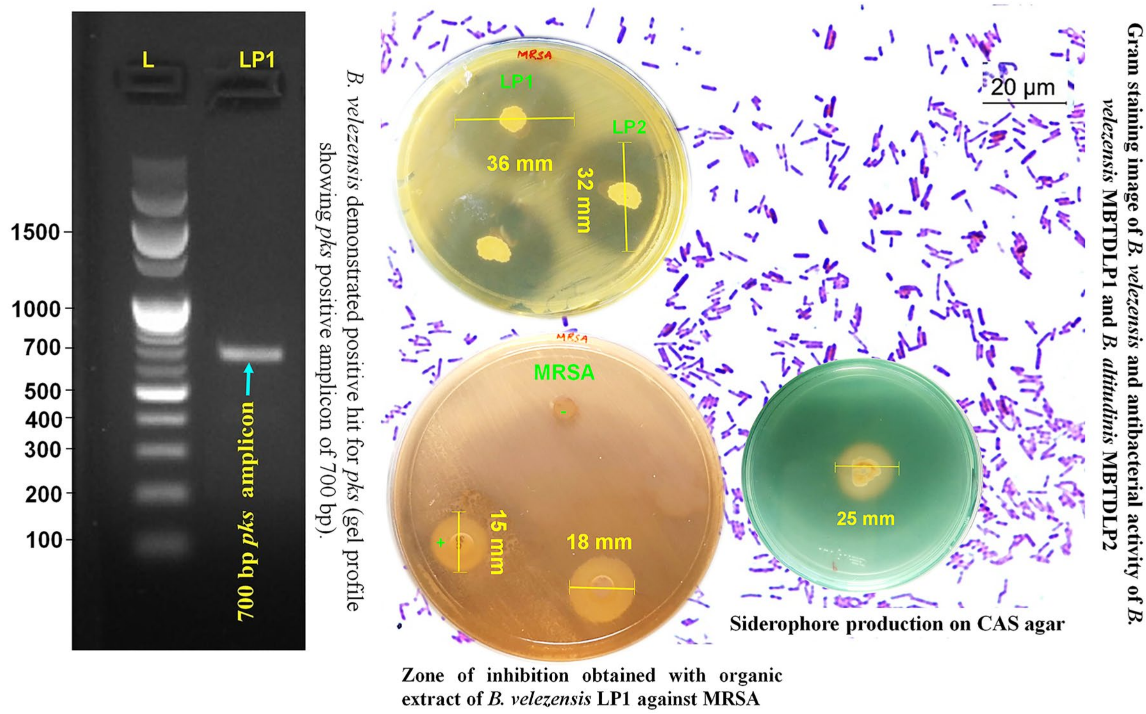
Graphic abstract



Keywords Marine macroalga-associated heterotrophic bacteria · *Bacillus velezensis* MBTDLP1 MTCC 13048 · Antimicrobial agent · Macrocyclic polyketide · Iron-chelating activities

Introduction

Marine macroalgae or seaweeds are sessile multicellular photosynthetic eukaryotes with specialized tissues as root system or vascular structures, and were found to harbor heterotrophic microbiome, which play an important functional role to induce innate resistance to the host organism against pathogenic microorganisms' abundance in oceanic waters (Egan et al. 2013). Over the past few decades, a large amount of research work was focused to isolate and characterize heterotrophic bacteria allied with marine algae, with the aim of understanding the structure, sequence, and community dynamics in the context of chemical ecology of macroalga–bacterial interface (Penesyan et al. 2009). Recently, studies involving algae, sponges and their associated bacteria unveiled that bioactive metabolites extracted from them has exceptional antibacterial and antioxidant activities, and related therapeutic applicability could be exploited in drug discovery (El-Moneam et al. 2017). Owing to the ability to occupy a variety of unique biological niches in an extremely challenging environment, marine heterotrophic microorganisms are considered an important source of bioactive natural products (Villarreal-Gómez et al. 2010) that act against an array of pathogens and making them use for biotechnological applications (Penesyan et al. 2009). The decreased efficacy of conventional antibiotic agents against the nosocomial pathogens due to their antibiotic resistance has paved way for the need for newer and more effective antimicrobial agents from unconventional natural habitats (Liu et al. 2019). With the ever-growing demand for discovery of novel antimicrobial agents to combat with the emerging superbugs, marine macroalgae-associated heterotrophic bacteria have emerged as novel sources of new antibacterial metabolites (Zheng et al. 2005; Kanagasabhapathy et al. 2008; Penesyan et al. 2009; Wiese et al. 2009; Chakraborty et al. 2017, 2020). Imamura et al. (1997) reported that a strain of *Pelagiobacter variabilis* isolated from the brown marine macro algae, *Pocockiella variegata* produced a phenazine molecule with potential antibacterial activity. Jamal et al. (2006) showed the antibacterial activity of *Bacillus licheniformis* against methicillin-resistant *Staphylococcus aureus* (MRSA) and vancomycin-resistant *Enterococcus faecalis* (VRE*fs*) was linked with the intertidal macroalga, *Fucus serratus* (2006). Symbiotic bacteria have a high potential to produce bioactive metabolites, which limit the growth of drug-resistant pathogens (Kizhakkekalam and Chakraborty 2019).

Surface-dwelling microbes were reported to adopt strategies to avoid colonization of potential competitors because of a highly competitive environment as well as limitations of resource and space (Egan et al. 2008).

Bioactive compounds were biosynthesized in response to the competition and predator deterrence in the heterotrophic bacteria associated with marine macroalgae. Notably, the secondary metabolites produced by macroalgae-associated microbiome could be of immense biomedical significance (Villarreal-Gómez et al. 2010). These specialized metabolites require mainly three classes of enzymes for their synthesis, non-ribosomal peptide synthetetases (*nrps*), polyketide synthases (*pks*), and *pks*/*nrps* hybrids from chimeric genes yielding chimeric products (Theobald et al. 2019; Andryukov et al. 2019). Interestingly, several antibiotics, such as tetracyclin and erythromycin are the products of *pks*-*nrps* biosynthetic gene clusters (Brinkmann et al. 2017). These studies demonstrated that the polyketide compounds reported from macroalgae were in fact the compounds from their associated bacterial community (Chakraborty et al. 2017). With reference to that of the earlier literature, it could be inferred that the bacterial communities associated with marine algae were repository of potential antibacterial agents with polyketide classes of chemistry (Kizhakkekalam and Chakraborty 2020) along with other class of bioactive compounds (Sun et al. 2019). The order *Bacillales* has its place to phylum *Firmicutes*, and the genera in this order was reported to exemplify a bountiful resource for diverse secondary metabolite gene assemblages. A recently studied complete genome sequencing analysis showed that greater than 30% of the *Firmicutes* were anticipated to shelter *pks*-*nrps* metabolite gene clusters (Aleti et al. 2015). Particularly, the genome sequence of widely distributed *Bacillus* spp. revealed that lesser than 5–10% of genome was implicated in antibiotic biosynthesis (Chen et al. 2009; Stein 2005).

Contribution of siderophore in imparting antimicrobial property has been the area of interest lately. Marine *Bacillus* were reported to produce antimicrobial compounds with novel modes of action to inhibit the pathogenic organisms (Blin et al. 2019), and biosynthesize siderophore type of metabolites that augment the transport of complexed Fe^{3+} into the bacterial cell (Rütschlin et al. 2017). To survive in iron-deficient environment, the microorganisms solubilize the insoluble ferric form of iron {Fe (III)} by small molecular, iron-chelating molecules called siderophores (Ribeiro and Simões 2019). Siderophores are crucial in antimicrobial approach owing to their importance in cell survival and pathogenicity. The iron chelators could competitively inhibit the pathogenic bacteria through iron starvation, thus limiting their growth (Ribeiro and Simões 2019). Simultaneously, Achard et al. 2013 reviewed that effective Fe (III) chelation of siderophores prevented the oxidative stress catalyzed by superoxide and Fe (III). Marine bacteria produce these Fe-transmission agents to compete for the restricted nutrient element that intercede the transport of Fe into the cytosol by explicit receptors and transport organizations (Rütschlin et al. 2017).

Increasing threat of the human and food-borne pathogens in the current scenario with evolved antibiotic resistance has led to increased exploitation of bioactives from various natural origins. Marine macroalgae act as potential host organisms for symbiotic existence for associated heterotrophic bacteria, which could be the promising sources of novel antimicrobial metabolites. In the background, the present work aimed to isolate the bacterial inhabitants associated with the intertidal macroalgae corresponding to *Phaeophyceae* and *Rhodophyceae* to assess their bioactive properties against selected pathogens, such as MRSA. The work also envisaged to characterize the selected bioactive isolates *Bacillus velezensis* MBTDLP1 and *Bacillus altitudinis* MBTDLP2, and analyzing the functional *nrps* and *pks-I* genes. Further objectives of this study were to assess the antibacterial activity of the organic extract from the most active bacterium *B. velezensis* MBTDLP1 coupled with its ability of siderophore production, characteristic spectral fingerprint analysis of the organic extract of *B. velezensis* and bioactivity-guided isolation of an antibacterial macrocyclic polyketide compound.

Materials and methods

Separation of marine macroalgae-associated heterotrophic bacteria

Marine macroalgal samples were collected from the intertidal region of Mandapam (9°17′N, 79°7′E) along the southeast coast of Indian peninsular, during low tide. Macroalgae associated with the family *Chlorophyceae*, *Rhodophyceae* and *Phaeophyceae* were collected by hand picking, and were used to isolate the macroalga-associated heterotrophic bacteria, as described earlier (Supplementary material S1) (Lemos et al. 1985).

Screening of bacterial isolates on the basis of antimicrobial activities and their characterization

The experimental pathogens utilized in the framework of this study were *Vibrio parahaemolyticus* (MTCC 451), *Vibrio parahaemolyticus* (ATCC 17802), *Aeromonas salmonicida* (ATCC 27013), *Yersinia enterocolitica* (MTCC 859), *Streptococcus pyogenes* (MTCC 1924), *Escherichia coli* (MTCC 443), *Edwardsiella tarda* (MTCC 2400), *Aeromonas caviae* (ATCC 15468), and methicillin-resistant *Staphylococcus aureus* (ATCC 33952) (MRSA). Initial classification of bioactive bacteria was performed by the spot-over-lawn antibacterial assay, and measured as the diameter of clearing zone in the growth of pathogen, as described previously (Kizhakkekalam and Chakraborty 2019). The bacterial isolates possessing significant antibacterial activities were identified by biochemical and microbiological characterization (Thilakan et al. 2016; Kreig and Holt 1984). Morphology of the bacterial colony was evaluated, along with Gram staining, citrate utilization, Voges–Proskauer, hydrolysis of gelatin/starch, reduction of nitrate/indole, and growth at different NaCl concentrations/temperatures. The identity of the strains was performed by 16S rRNA gene sequencing followed by phylogenetic analysis using BLAST tool for the similarity comparison among the strains. Phylogenetic tree was created by maximum likelihood method (conducted on MEGA X). Preliminary tree(s) for the empirical search were acquired by utilizing Neighbor-Joining and BioNJ algorithms to the matrices of pair-wise spaces appraised with Tamura–Nei model, and thenceforth choosing the topology with greater value of log-likelihood. The tree was sketched to scale, using lengths of the branches computed in the number of substitutions for each location. GenElute isolation kit (Sigma-Aldrich, MO, USA) was utilized to extract bacterial genomic DNA, which was enumerated with a biophotometer (Eppendorf, Germany). The biotyping was corroborated by matrix-assisted laser desorption/ionization–time of flight mass spectral analysis (MALDI–ToF) (Dieckmann et al. 2005; Ghyselinck et al. 2011; Spitaels et al. 2016) and biolog characterization (Duarte et al. 2019). The primer sequences used for the PCR reaction and the reaction conditions are presented in Table 1. PCR was accomplished in the reaction mixture of PCR 2X master mix (12.5 µL, TAKARA), DNA (1 ng), forward/reverse primer (1 µL each), and volume made up to 25 µL using sterile water supplied with the kit. PCR specifications were as follows (Biorad, CA, USA): primary denaturation (94 °C/5 min) and following 35 cycles (94 °C/1 min, 58 °C/1 min, 72 °C/2 min), followed by extension (72 °C/5 min). The amplified PCR products were subjected to molecular size determination by equating with 1 kb ladder (Thilakan et al. 2016) before being sequenced and banked (in GenBank). Sequence data were weighed against the reference sequences by means of BLAST program, and was phylogenetically analyzed. Partial 16S rRNA gene sequences of two bacterial isolates were submitted in the NCBI GenBank (accession numbers of MT122835, MT122905). These bioactive isolates were deposited in the Microbial Type Culture Collection (MTCC), an International microbial depository authority in the South-East Asia under the Budapest Treaty, with accession numbers of *B. velezensis* MTCC 13048 and *B. altitudinis* MTCC 13044, respectively.

Screening of *pks*-I and *nrps* genes

The presence of *pks*-I and *nrps* genes implicated in the biosynthesis of bioactive metabolites in the most active strain was amplified with different sets of degenerate primers (Table 1). PCR was accomplished in the reaction mixture detailed in the previous section, whereas the PCR specifications were as follows: primary and secondary denaturation (94 °C/5 min and 35 cycles at 95 °C/1 min, respectively), annealing (45 °C/1 min for *pks*; 55 °C/1 min for *nrps*), and extension (72 °C/5 min). The amplified PCR products were subjected to agarose gel electrophoretic analysis, and 700/800 and 1000/1400 bp fragments were contemplated as *pks-I* and *nrps* gene products, respectively. The *pks-I*/*nrps* products were sequenced before being deposited to the NCBI GenBank. The partial *pks* sequences corresponding

Table 1 Polymerase chain reaction primers used for 16S rRNA and *pks, nrps* gene amplification

Primer	Target	Sequence (5′–3′)	References
16S1[a]	*16S rRNA*	GAGTTTGATCCTGGCTCA	Xia et al. (2015)
16S2[a]	*16S rRNA*	ACGGCTACCTTGTTACGACTT	Xia et al. (2015)
GCF	*pks*	GCSATGGAYCCSCARCARCGSVT	Schirmer et al. (2005)
GCR	*pks*	GTSCCSGTSCRTGSSCYTCSAC	Schirmer et al. (2005)
GBF	*pks*	RTRGAYCCNCAGCAICG	Zhang et al. 2009
GBR	*pks*	VGTNCCNGTGCCRTG	Zhang et al. (2009)
KS11F	*pks*	GCIATGGAYCCICARCARMGIVT	Schirmer et al. (2005)
KS12R	*pks*	GTICCIGTICCRTGISCYTCIAC	Schirmer et al. (2005)
KSDPQQF	*pks*	MGNGARGARGCNNWNSMNATGGAYCCN-CARCANMG	Zhang et al. (2009)
KSHSGDR	*pks*	GGRTCNCCNARNSWNGTNCCNGTNCCRTG	Zhang et al. (2009)
MTF	*nrps*	GCNGGYGGYGCNTAYGTNCC	Zhao et al. (2008)
MTR	*nrps*	CCNCGDATYTTNACYTG	Zhao et al. (2008)

Different sets of oligonucleotide primers were used to amplify the partial region specific for *pks* and *nrps* genes. The resulting amplicons of size 700 bp corresponds to *pks-I* specific genes, while 1000–1500 bp amplicons gave + results for *nrps* genes

[a]Universal primers (commonly used in identifying the taxonomic position of prokaryotic species)

to *B. velezensis* MBTDLP1 was deposited under the accession number MT394492. Phylogenetic examination of the *pks* gene sequence obtained was performed, and a tree was assembled by maximum likelihood method using MEGA X. Preliminary tree(s) for the empirical search were acquired by utilizing the maximum parsimony method, and the tree was sketched to scale, using lengths of the branches computed in the number of substitutions for each location. The proportion of sites where at least 1 unambiguous base was present in at least 1 sequence for each descendent clade was shown next to each internal node in the tree.

Antibiotic susceptibility and siderophore detection assay

Antibiotic sensitivity of the selected bacterium *B. velezensis* MBTDLP1 selected by spot-over-lawn assay and *pks*-I screening was ascertained using antibiotic-infused octadiscs (HiMedia Laboratories LLC, PA, USA), as described earlier (Kizhakkekalam et al. 2020). The latter comprised of eight antibiotics, such as cephalothin, clindamycin, ofloxacin, trimoxazole, penicillin-G, gentamicin, erythromycin, and vancomycin, which were positioned on the center of the MHA plate, as described earlier (Kizhakkekalam et al. 2020). The bacterial strain was further checked for assembly of siderophore by universal chrome azurol sulfonate (CAS) analysis. Concisely, CAS (121 mg) was dispelled in distilled water (~ 100 mL) and ferric chloride ($FeCl_3 \cdot 6H_2O$, 20 mL of 1 mM) solution was made in an acid (HCl, 1×10^{-3} M) before being put in cetyltrimethylammonium bromide ($C_{19}H_{42}N^+Br^-$, CTAB 20 mL, 729 mg CTAB/0.4 L distilled water) under stirring followed by sterilization. Agar plates containing CAS were set by incorporation of CAS reagent (100 mL) into the CAS agar medium (sterilized, 860 mL) followed by addition of casamino acid (30 mL) along with sterile glucose solution (10 mL) onto the media before being solidified. The bacterial strain was spot-inoculated on the CAS plate before being incubated at 37 °C for one week, and perceived for the development of yellow–orange area about the bacterial colony (Louden et al. 2011). The plate showed positive yellow zone of diameter 25 mm depicting the production of siderophore. Siderophore biosynthetic genes were identified in the genome of candidate bacterium, and the sequences were submitted in the NCBI GenBank under the accession numbers MT682354 and MT682353. Whole genome sequence of the isolate *B. velezensis* MBTDLP1 was deposited under the accession number JABZEG000000000 in the NCBI database. Siderophore biosynthetic genes were identified in the genome of *B. velezensis* MBTDLP1, and the sequences were submitted in the NCBI GenBank under the accession numbers MT682354 and MT682353. Phylogenetic tree was created by maximum likelihood method (conducted on MEGA X) with siderophore biosynthetic genes predicted from bacterial genome (NCBI accession numbers of MT682354 and MT682353). Preliminary tree(s) for the empirical search were obtained by applying the maximum parsimony method, and the tree was sketched to scale, using lengths of the branches figured in the number of substitutions for every position. The fraction of situates where no less than 1 explicit base was present in at least 1 sequence for every descendent clade was displayed subsequent to every single internal node in the tree.

Bacterial crude extract preparation for antibacterial and ferrous ion chelation assays

Extracellular metabolites of the bacterium *B. velezensis* MBTDLP1 (MTCC 13048) possess prospective antimicrobial properties against pathogens, was used to develop bacterial extract with solvent ethyl acetate, and assayed for antibacterial potential by disc diffusion method, MIC (minimum inhibitory concentration), and MBC (minimum bactericidal concentration) (Kizhakkekalam and Chakraborty 2019; Bauer et al. 1966) against clinically recognized pathogens (Supplementary material S2). Iron chelation potential of the solvent extract of *B. velezensis* MBTDLP1 was studied using ferrous ion (Fe^{2+}) chelating assay (Gülçin 2007) (Supplementary material S3). Siderophore production was detected by the formation of yellow orange zone on (chrome azurol sulfonate) CAS agar, using spot inoculation of *B. velezensis* and *B. altitudinis* (shown in Fig. 3C). The concentration of bacterial extract necessary to achieve 50% ferrous ion chelating ability was calculated as IC_{50} (effective concentration).

Spectroscopic analyses and preparatory chromatography

Spectroscopic thumbprint features of the solvent extract of *B. velezensis* MBTDLP1 (MTCC 13048) was analyzed by ^{1}H-NMR (proton-nuclear magnetic resonance) study (Bruker Avance-500 MHz, Billerica, MA, USA). Protons at the characteristic domain of the ^{1}H-NMR spectrum were summed up to ascertain their cumulative number (designated as proton integrals, ΣH), and chemical shift (δ_H) values were expressed as parts per million (δ_H, ppm). The bacterial extract (~ 2.5 g) was separated over an octadecylsilane reverse-phase (RP-C_{18} stationary-phase, 25 × 0.46 cm, 5 µm) column (using MeOH: H_2O, 8:2 v/v, isocratic) mounted on a preparatory HPLC (high-performance liquid chromatograph; Shimadzu, Nakagyo-Ku, Japan) system coupled with a diode-array detector to yield a homogenous fraction (~ 90 mg) with ~ 98% purity. The titled compound isolated from *B. velezensis* were evaluated for antibacterial activity by disc diffusion method on Muller Hinton Agar plates with overnight grown cultures of test pathogens (10^8 CFU/mL; (CFU colony forming unit) and micro-dilution method (expressed as minimum inhibitory concentration MIC, μg mL^{-1}) against clinically recognized pathogenic bacteria. Nuclear magnetic resonance (NMR) spectra (500 MHz; Bruker Advance DPX 500, Karlsruhe, Germany), 1D (^{1}H-NMR and ^{13}C-NMR) and 2D (correlation spectroscopy ^{1}H–^{1}H COSY and heteronuclear multiple-bond correlation spectroscopy HMBC). High-resolution electrospray ionization mass spectral (HRESIMS) analysis was performed in ESI positive mode on Agilent 6520 accurate mass Q-TOF LC/MS (Agilent, Santa Clara, CA, USA) coupled with a HPLC (Agilent LC 1200) equipped with a reverse-phase C_{18} column (Extend™, 1.8 μm film thickness, 2.1 × 50 mm).

Data analysis

Statistical Program for Social Sciences (SPSS Inc, CA, USA; ver. 10.0) was employed for the data analysis. The experiments were performed in triplicate, whereas the means of all parameters were measured for significance ($p \leq 0.05$) by ANOVA (analysis of variance).

Results

Isolation of marine algae associated heterotrophic bacteria and antibacterial activity screening

Marine algae have its place in the families of *Phaeophyceae* and *Rhodophyceae*, which are ubiquitous alongside Southeast coast of India, were collected, and the bacteria associated with them were purified to homogeneity and assessed for antimicrobial potentials. Forty bacterial strains were isolated from eight different macroalgae belonging to brown algae, *Sargassum wightii, Turbinaria ornata, Padina pavonica, Sargassum tenerrimum, Sargassum marginatum* and red algae, *Laurencia papillosa, Gracilaria edulis* and *Gracilaria salicornia.* A total of 20% of the isolates was contributed by *S. wightii,* followed by *T. ornata, P. pavonica, S. tenerrimum* and *S. marginatum,* each contributed 12.5% of the total isolates. The marine algae *L. papillosa* and *G. salicornia* contributed to about 10% of the cumulative isolates, while *G. edulis* was found to be associated with remaining isolates (Fig. 1A). From the 40 isolates, two strains showed significant antibacterial activities collectively against all the test pathogens (Table 2). Amongst the isolates, *B. subtilis* occupied the major share of bioactive isolates (33% of the total isolates) followed by *B. velezensis* (20%) (Fig. 1B), and those could able to endure sub-culturing under the laboratory culture conditions. Out of the total bacterial isolates purified to homogeneity, eight of them exhibited prospective antimicrobial activity. The isolates MBTDLP1 *B. velezensis* MTCC13048 and MBTDLP2 *B. altituidnis* MTCC13044 isolated from *L. papillosa* exhibited greater antimicrobial properties with the inhibition zones of 36 and 32 mm, respectively, against MRSA (Fig. 2A).

Identification of bioactive heterotrophic isolates

The bacteria displaying potential antimicrobial properties against test pathogens were identified by means of microbiological, physiological, and biochemical tests, as described previously (Thilakan et al. 2016). The white, undulated colonies of the isolates LP1 and LP2 were Gram positive with short-chained spores (Table S1). Identity of the representative isolates from *L. papillosa* was further affirmed with 16S rRNA gene analysis, MALDI–ToF, and biolog characterization. The isolates were converged onto *Bacillus* genus of the phylum *Firmicutes,* by biochemical tests and into species level identification by 16S rRNA gene sequencing (Fig. S1) that resulted in *Bacillus velezensis* MBTDLP1 and *Bacillus altitudinis* MBTDLP2. The whole genome of strain *B. velezensis* MBTDLP1 was sequenced (accession number JABZEG000000000 in the NCBI database), which further substantiated the identity of the bacterium. The bacterial strain was deposited in the Budapest Treaty recognized international microbial repository as *B. velezensis* MTCC13048 and *B. altitudinis* MTCC13044. The 16S rRNAs sequence was equated with their neighboring GenBank lineages, and a phylogenetic tree was built (Fig. 2B). Evolutionary lineage was surmised by maximum likelihood method and Tamura–Nei model (Tamura and Nei 1993), and the tree with highest log probability (− 4100.03) was displayed.

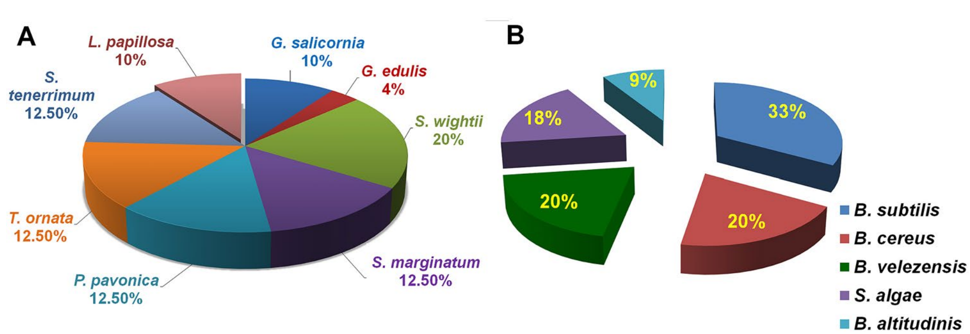


Fig. 1 **A** Distribution of marine macroalgal samples collected along southeast coast of India. **B** Distribution of bioactive isolates associated with marine algae was exemplified as percentage contribution

Table 2 Antibacterial activity of the marine macroalgae-associated heterotrophic bacterial isolates against test pathogens by spot-over-lawn assay

Macroalgae species	Marine macroalgae-associated bacterial strains[a]	Macroalgae associated *Bacillus* species	Antimicrobial activity against test pathogens [inhibition zone diameter (in mm)][b]							
			V. parahaemolyticus MTCC 451	*A. salmonicida* ATCC 27013	*Y. enterocolitica* MTCC 859	*E. coli* MTCC 443	*S. pyogenes* MTCC 1924	*E. tarda* MTCC 2400	*A. caviae* ATCC 15468	MRSA ATCC 33592
S. marginatum	SM1[c]	*B. amyloliquefaciens*	7.3 ± 0.47	ND	12.6 ± 0.43	7.0 ± 0.24	ND	8.0 ± 0.24	ND	ND
S. wightii	SW1[d]	*B. subtilis*	ND	ND	17.6 ± 0.43	ND	ND	8.0 ± 0.16	ND	ND
T. ornata	TO2[e]	*B. safensis*	ND	ND	14.5 ± 0.40	7.0 ± 0.16	ND	ND	ND	ND
L. papillosa	LP1[f]	*B. velezensis*	35 ± 0.81	7.0 ± 0.29	20 ± 0.81	12 ± 0.23	20.3 ± 0.62	12 ± 0.40	10 ± 0.81	36 ± 0.47
L. papillosa	LP2[f]	*B. altituidnis*	17 ± 0.81	15 ± 0.84	11 ± 0.61	7.0 ± 0.32	18 ± 0.81	11 ± 0.62	7.0 ± 0.4	32 ± 0.29
G. edulis	GE1[g]	*B. subtilis*	ND	ND	ND	7.0 ± 0.47	ND	ND	ND	ND
S. tenerrimum	ST1[h]	*B. amyloliquefaciens*	7.6 ± 0.47	ND	ND	7.0 ± 0.32	ND	ND	ND	ND
S. tenerrimum	ST2[h]	*B. safensis*	ND	ND	12.5 ± 0.40	7.5 ± 0.40	ND	8.0 ± 0.62	ND	ND

ND Not detected any significant inhibition

[a]NCBI GenBank accession numbers were shown in parentheses

[b]Samples were analyzed in triplicates ($n = 3$) and expressed as mean ± standard deviation

[c]SM—*Sargassum marginatum*

[d]SW—*Sargassum wightii*

[e]TO—*Turbinaria ornate*

[f]LP—*Laurencia papillosa*

[g]GE—*Gracilaria edulis*

[h]ST—*Sargassum tenerrimum*

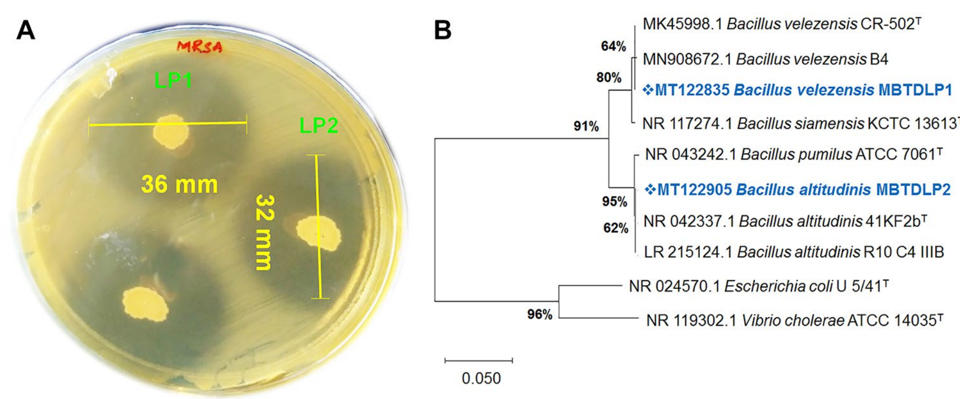


Fig. 2 **A** Antibacterial activity exhibited by the bacterial isolates *B. velezensis* MBTDLP1 (MTCC 13048) and *B. altitudinis* MBTDLP2 (MTCC 13044) against MRSA as visualized on Mueller Hinton agar plates on spot-on-lawn assay. **B** Phylogenetic tree from the partial 16S rRNA sequences between the isolates and other related species. The isolates were categorized as *B. velezensis* (LP1) and *B. altitudinis* (LP2)

This analysis encompassed 10 nucleotide sequences, and 1513 loci in the dataset, whereas evolutionary assessments were effected in MEGA X (Kumar et al. 2018). The selected bacterial isolate was found susceptible to the commercially available antibiotics (Fig. S2).

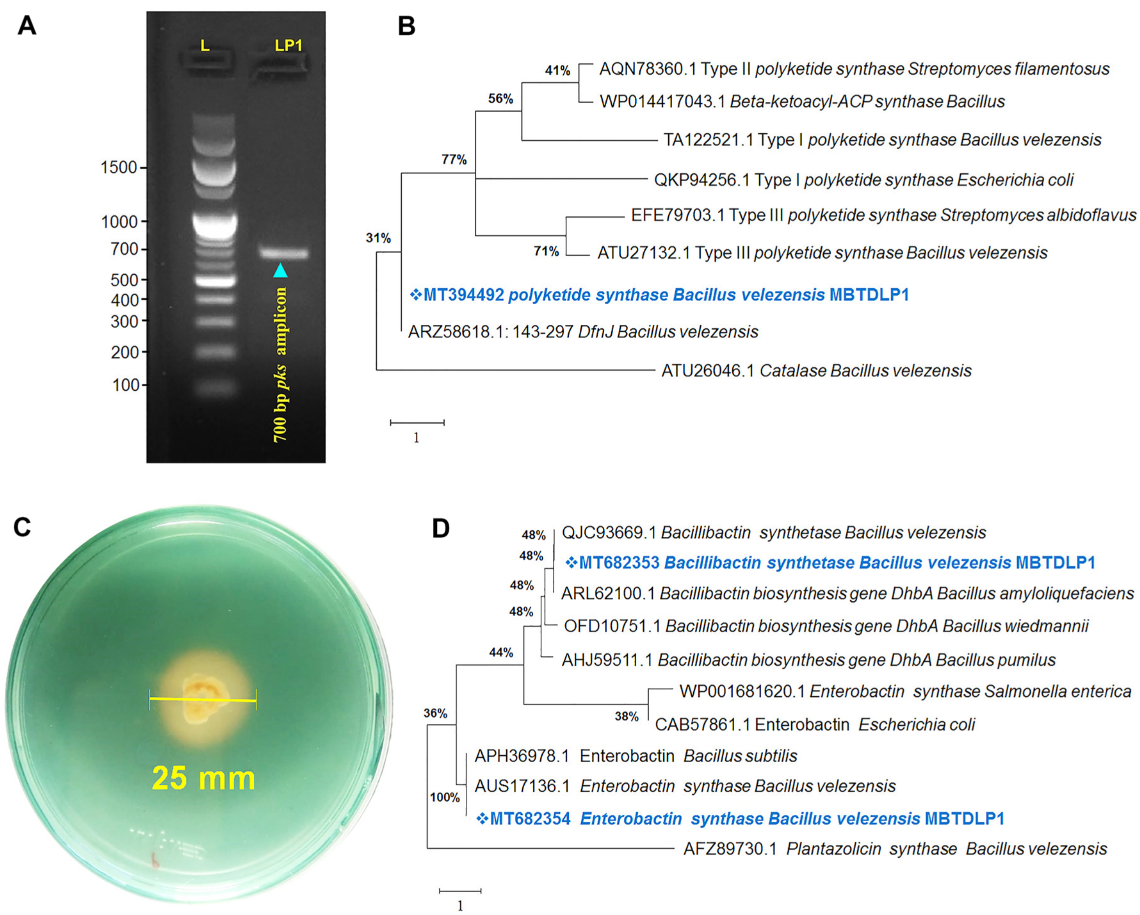


Fig. 3 *B. velezensis* MBTDLP1 (MTCC 13048) demonstrated positive hit for polyketide synthase gene (*pks*) and the amplified type-I *pks* gene array was banked (NCBI GenBank accession number MT394492). **A** Gel profile showing the *pks* specific analysis with positive amplicon of 700 bp. Lane LP1, *pks*-product (~700 bp) and Lane L, molecular marker (Gene Ruler™: 0.1–1.5 kb DNA ladder, Thermo Scientific, MA, USA). **B** Phylogenetic tree for *pks* gene sequence. **C** Siderophore production on chrome azurol sulfonate (CAS) agar for *B. velezensis* MBTDLP1 (MTCC 13048). **D** Phylogenetic tree for siderophore biosynthetic genes predicted from bacterial genome (NCBI accession numbers of MT682354 and MT682353)

Type-1 *pks* gene amplification

Type-1 *pks* gene was amplified from *B. velezensis* MBT-DLP1 (MTCC13048) (Fig. 3A), and the sequence showing significant homology in BLAST search was submitted to the NCBI GenBank (accession number MT394492) (Fig. 3B). Functional *pks* gene amplification of the isolates with various primers (GCF/GCR, KS11F/KS12R, KSD-PQQF/KSHSGDR, GBF/GBR) as well as *nrps* gene amplification by the primer MTF/MTR were performed (Fig. S3). Evolutionary lineage was surmised by maximum likelihood technique and JTT matrix-centered model (Jones et al. 1992), and the tree with highest log probability (− 7532.22) was displayed. This analysis encompassed 9 amino acid sequences, and 493 loci in the dataset, whereas evolutionary assessments were effected in MEGA X (Kumar et al. 2018). No specific amplification was detected for the *nrps* primers used. The phylogeny of the sequenced KS domains of *B. velezensis* MBTDLP1 and amino acid adenylation domain-containing protein, categorized the presumed amino acid sequences as bacterial type-I *pks*.

Siderophore production and ferrous ion chelation by *B. velezensis* MBTDLP1 (MTCC 13048)

Siderophore production was marked by a yellow zone of diameter (25 mm) using CAS agar detection (Fig. 3C). Presence of the genes corresponding to siderophore biosynthesis was predicted and annotated in the bacterial genome. One of the genes of 1626 bp (NCBI GenBank accession number MT682354) could translate to (2,3 dihydroxybenzoyl) adenylate synthase, and other gene of 786 bp (MT682353) translated to 2,3-dihydro-2,3-dihydroxybenzoate-NAD^+oxidoreductase. Evolutionary lineage was deduced by maximum likelihood method and JTT matrix-centered model (Jones et al. 1992), and the tree with highest log probability (− 4807.58) was presented. This analysis comprised 11 amino acid sequences, and 541 loci in the dataset, while evolutionary assessments were realized in MEGA X (Fig. 3D). Ferrous ion chelation ability of the extracellular extract of MBTDLP1 *B. velezensis* (MTCC 13048) exhibited the IC_{50} value of 4.2 mg/mL.

Antimicrobial properties of the organic extract of *B. velezensis* MBTDLP1

Bacterial growth kinetics deduced that *B. velezensis* MBT-DLP1 took 12 h to adapt to logarithmic phase and exponentially divided up to 56 h (Fig. 4A). During the stationary phase (56–64 h), cells appeared to switch to survival mode, and enable the production of secondary metabolites for survival. Antagonist activity of *B. velezensis* was also increased during 56–60 h of growth. Bacterial growth kinetics of *B. velezensis* MBTDLP1 against the expression of bioactivity in spot-over-lawn assay showed that antibacterial activity reached at its peak (35 mm inhibition zone against MRSA) after 56 h of incubation (Fig. 4A). Thus, the extracellular metabolites of *B. velezensis* was retrieved by surface culturing and incubating the cells for 60 h over nutrient agar and extracted using ethyl acetate to bring in the organic extract (5 g). The antimicrobial properties of the organic extract showed 28 mm of inhibition zone against *V. parahemolyticus* ATCC 17802, 21 mm against *S. pyogenes* MTCC 1924, and 18 mm against MRSA ATCC 33952, in descending order (Fig. 4B–D) in comparison to the positive control, chloramphenicol with 15 mm zone of inhibition against MRSA, 11 mm against *V. parahemolyticus* and no zone against *S. pyogenes*. It was perceived that the MIC of the organic extract of *B. velezensis* was 7.5 µg/mL against *V. parahemolyticus* and *S. pyogenes*, and 15 µg/mL against MRSA. The antibacterial activity of the organic extract assessed by disc diffusion assay and minimum inhibition concentration (by micro-dilution method) are summarized in the Table 3.

Spectroscopic analysis and antimicrobial activity

Labeling of ^{1}H-NMR protons of the functional groups in the ethyl acetate extract of *B. velezensis* MTCC13048 (Fig. S4) showed that aromatic proton signals appeared at δ_H 6.5–8.5 with the proton integral ($\sum H$) of about 17.1, whereas those at δ_H 2–3 ($\sum H$ 6.9) could recognize the allylic/acetyl groups. Proton integrals at the olefinic section of the ^{1}H-NMR spectrum (δ_H 4.5–6) was found to be considerably high ($\sum H$ 3.1). The organic extract of *B. velezensis* displayed greater proton integral owing to the preponderance of electronegative functional groups in the downfield section of the ^{1}H-NMR spectrum. Repeated HPLC fractionation of the crude organic extract over a RP-C_{18} column could result in the purification of a homogenous fraction {macrocyclic polyketide R_t C_{18}-RPHPLC, methanol–acetonitrile, 1:1 v/v 2.675 min; 8-(pent-2-enyl)-1-oxo-5a,8a-dioxacyclododecanyl-3-oxyethyl-5b′-methyl-5′-(7′(9′(methoxyethyl)-dihydrofuranyl) propanamido) succinate (Fig. 4E, F), BV-1, *m/z* 681.4366 $[M+H]^+$ for $C_{41}H_{60}O_8$} having greater than 98% purity with promising antimicrobial properties (MIC 0.38 µg/mL) against MRSA ATCC 33592.

Spectroscopic data

^{1}H-NMR $_{CDCl3}$ (500 MHz, *J* in Hz): δ_H 2.71 (2H, d, 4.1), 3.73 (1H, m), 4.90 (1H, dd, 12.5, 3.1), 6.24 (1H, d, 13.2), 3.86 (2H, t, 4.8), 1.63 (2H, q, 6.5), 3.39 (1H, t, 6.1), 2.00 (2H, q, 5.0), 0.88 (3H, t, 6.1), 5.35 (1H, q, 3.9), 1.94 (3H, d, 3.5), 3.53 (2H, t, 6.1), 2.52 (2H, t, 5.9), 3.65 (2H, t, 6.5), 4.08 (2H, t, 6.4), 2.85 (2H, d, 5.50), 4.70 (1H, t, 4.4), 3.67 (3H, s), 8.01 (1H, d, 7.3), 2.13 (2H, t, 5.8), 2.30 (2H, t, 7.6),

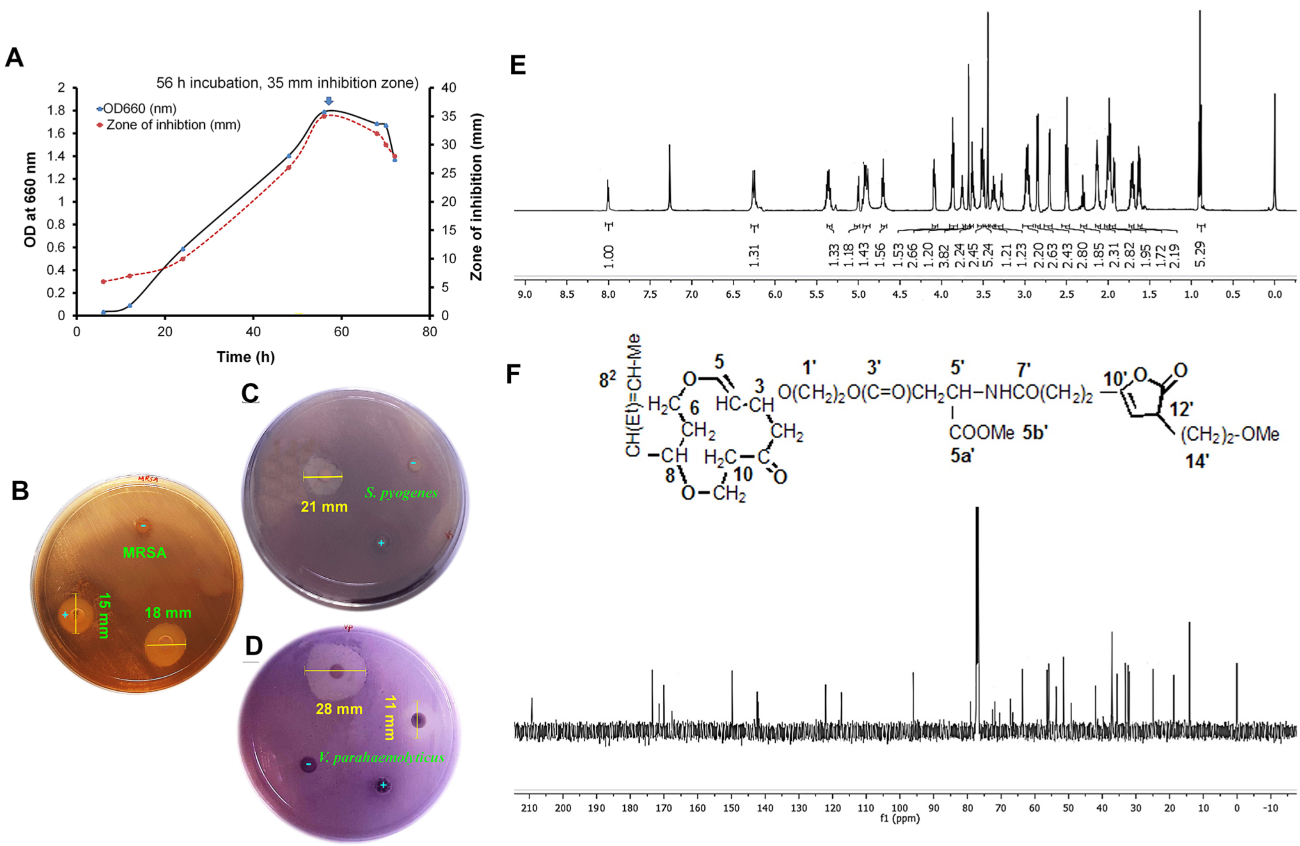


Fig. 4 **A** Bacterial growth kinetics of *B. velezensis* MBTDLP1 against expression of bioactivity against methicillin-resistant *Staphylococcus aureus* in spot-over-lawn assay. Disc diffusion assay results of *B. velezensis* MBTDLP1 organic extract, against (**B**) methicillin-resistant *S. aureus*, (**C**) *S. pyogenes* and (**D**) *V. parahaemolyticus*. Chloramphenicol (30 µg) and ethyl acetate, which were used as the positive and negative control, were denoted with (+) and (−), respectively. Stacked plot representing the (**E**) ^{1}H-NMR and (**F**) ^{13}C-NMR spectra of the polyketide-originated macrolactin

2.96 (2H, dt, 17.1, 5.7), 5.01 (1H, d, 3.6), 1.72 (2H, m), 3.32 (2H, t, 5.4), 3.46 (3H, s); ^{13}C-NMR $_{CDCl3}$ (125 MHz): δ_C 209.19 (C-1), 53.68 (C-2), 70.40 (C-3), 96.14 (C-4), 141.81 (C-5), 63.88 (C-6), 35.91 (C-7), 72.53 (C-8), 142.49 (C-8^1), 24.96 (C-8a^1), 14.15 (C-8b^1), 121.96 (C-8^2), 18.42 (C-8^3), 56.54 (C-9), 41.95 (C-10), 67.27 (C-1′), 66.54 (C-2′), 170.08 (C-3′), 37.11 (C-4′), 56.60 (C-5′), 171.28 (C-5a′), 51.49 (C-5b′), 173.87 (C-7′), 33.29 (C-8′), 31.79 (C-9′), 149.90 (C-10′), 167.26 (C-11′), 49.36 (C-12′), 116.93 (C-13′), 32.08 (C-14′), 71.91 (C-15′), 78.96 (C-16′); ^{1}H-^{13}C HMBC: H-2/C-1, 3; H-3/C-1; H-4/C-3, 5; H-5/C-6;H-7/C-6, 8; H-8/C-6, 9, 8^1; H-8a^1/C-8b^1; H-8b^1-C-8^1; H-8^2/C-8a^2; H-8^3/C-8^2; H-9/C-1; H-10/C-1, 9; H-1′/C-3, 2′; H-2′-C-3′; H-4′/C-3′, 5′; H-5′/C-3′, 5a′; H-5b′/C-5a′; H-8′/C-7′; H-9′/C-7′,10′; H-12'/C-10', 11',13'; H-14'/C-11'; H-15'/C-14'; H-16'/C-15'; HRESIMS *m/z* 638.3181 [M + H$^+$].

Discussion

The studies revolving marine free-living and symbiotic bacteria were propagated since 1950s after the discovery of their ability to produce antimicrobial metabolites, even though their occurrences were confirmed only in late 1990s (Vijayalakshmi et al. 2008; El-Shafay et al. 2016). At a recent time, heterotrophic bacterial groups in marine algae have fascinated the marine biotechnologists, on account of their prospects to biosynthesize different class of bioactive compounds with potential pharmacological connotations (Winter et al. 2013; Silva et al. 2020; Chakraborty et al. 2021). The bacterial classes have their places in *Firmicutes, Gammaproteobacteria* and *Bacteroidetes* were found to exist in relationship with marine algae (Wiese et al. 2009), and have prospective therapeutic properties.

In this study, the intertidal macroalgae belonging to the classes Rhodophyceae and Phaeophyceae were evaluated for the presence of associated heterotrophic bacteria. The overall distribution of microbial diversity demonstrated that

Table 3 Antibacterial activities of the organic extract of *B. velezensis* MBTDLP1 (MTCC 13048) against pathogenic bacteria

Test pathogens	Crude extract from *B. velezensis* MBTDLP1[a]		Chloramphenicol[c]	
	Inhibition zone diameter (mm)	MIC (MBC)[b]	Inhibition zone diameter (mm)	MIC (MBC)[b]
Methicillin resistant *Staphylococcus aureus* (ATCC 33592)	$18^{d} \pm 0.02$	15 (15)	$15^{d} \pm 0.20$	6.25 (6.25)
Vibrio parahaemolyticus (MTCC 451)	$28^{e} \pm 0.25$	7.5 (7.5)	$11^{e} \pm 0.11$	6.25 (10.0)
Vibrio parahaemolyticus (ATCC 17802)	$22^{f} \pm 0.10$	7.5 (15)	ND	6.25 (12.5)
Edwardsiella tarda (MTCC 2400)	$17^{d} \pm 0.12$	15 (15)	$8^{f} \pm 0.23$	6.25 (6.25)
Aeromonas caviae (MTCC 15468)	$15^{d} \pm 0.06$	15 (15)	$9^{f} \pm 0.32$	6.25 (6.25)
Aeromonas salmonicida (ATCC 27013)	$19^{f} \pm 0.20$	15 (15)	$10^{e} \pm 0.20$	12.5 (12.5)
Streptococcus pyogenes (MTCC 1924)	$21^{f} \pm 0.13$	7.5 (15)	ND	12.5 (12.5)
Escherichia coli (MTCC 443)	$12^{d} \pm 0.05$	15 (15)	$10^{e} \pm 0.23$	6.25 (6.25)
Yersinia enterocolitica (MTCC 859)	$17^{d} \pm 0.12$	15 (15)	$9^{f} \pm 0.22$	6.25 (6.25)

Ethyl acetate was used as negative control and commercially available antibiotic disc (HiMedia, PA, USA) as positive control

[a]30 µg/mL of crude extract

[b]MBC values of candidate bacterium and chloramphenicol were described in parentheses, and expressed asµg/mL

[c]30 µg/mL of crude extract

ND- non-detectable zone of clearance

[d−f] Column-wise values with different superscripts of this type indicate significant difference ($p < 0.05$), which implied for the statistical evaluation of the results. Triplicate values were taken and the variance analyses (ANOVA) were carried out (using Statistical Program for Social Sciences 13.0) for means of all parameters to examine the significance level ($p < 0.05$). Results were expressed as mean ± SD (n = 3)

a greater part of the isolates could fit in to the phylum *Gammaproteobacteria* (62%), and from that 18% of bioactive isolates was obtained. *Firmicutes* represented 38% of isolates from which majority of the bioactive isolates was retrieved (Fig. 1B). Earlier, the isolates from brown and red macroalgae were reported to possess antibacterial activities against several pathogens, including those, which were drug resistant, such as MRSA (Thilakan et al. 2016; Kizhakkekalam et al. 2020).

Heterotrophic isolates from the marine macroalgae were assessed for their antibacterial activities. Despite selecting the isolates displaying inhibition against any one of the test pathogens, the most active isolates, which showed wide-spectrum of inhibitory activity against the clinical pathogens, were chosen for further studies (Fig. 2A; Table 2).

The strains *B. velezensis* MBTDLP1 and *B. altitudinis* MBTDLP2 belonging to the family *Firmicutes*, isolated from the red algae *L. papillosa*, showed significant antagonistic properties against pathogens of human and food-borne diseases, including MRSA, with zones of inhibition of 36 and 32 mm, respectively. These bacteria were identified with standard biochemical methods, biolog characterization and MALDI biotyping followed by 16s rRNA sequencing before being deposited in the Microbial Type Culture Collection and Gene Bank (MTCC) as *B. velezensis* MTCC 13048 and *B. altitudinis* MTCC 13044. Noticeably, the bacteria also showed significant susceptibility to the commercial available antibiotics (Fig. S2), and thus could disregard the likelihood of pathogenicity. Notably, among *Firmicutes* group, *Bacillus* species were dominantly present on the surface of diverse marine algae, and were acknowledged as vital sources of bioactive agents with biomedical significance (Goecke et al. 2010). *Bacillus* species of marine origin were accounted for to produce promising antimicrobial bioactive metabolites, such as bacteriocins, polyketides, and other related compounds (Goecke et al. 2010). Non-ribosomal peptide synthetase (*nrps*), polyketide synthase (*pks*), and *nrps–pks* hybrid were found to catalyze the biosynthesis of small molecular weight peptides, polyketides, and compounds with hybrid polyketide-peptide functionalities, respectively (Wang et al. 2014). Herein, *pks* gene amplified from *B. velezensis* MBTDLP1 (MT394492) (Fig. 3A) showed resemblance to type-I *pks* of *B. velezensis* in the BLAST similarity search (Fig. 3B). Marine macroalga-associated *B. subtilis* with *pks*-1 gene, showed a new variant of polyketide furanoterpenoids with potential activity against food-borne pathogens (Chakraborty et al. 2017). A rare chemistry of polyketide-spanned macrolides showing potential growth inhibition on drug-resistant bacteria were isolated from *H. valentiae*-associated *B. amyloliquefaciens* (Kizhakkekalam et al. 2020). Marine originated *Bacillus* sp. were reported to produce promising antimicrobial metabolites, such as bacteriocins, polyketides etc. (Goecke et al. 2010).

Most of the bacteria secrete strong iron-chelating secondary metabolites, such as siderophores, which could quench iron from their surroundings (Kramer et al. 2020) to establish a competition between bacterial species for iron acquisition. This ability of bacteria could contribute to the competitive characteristics enabling their antagonist activity against pathogenic bacteria. Siderophores are typically synthetized by *nrps* or *pks* domains, which work in coordination with *nrps* modules (Kramer et al. 2020). The present study reported the production of siderophore from *B. velezensis* MBTDLP1 on the CAS agar plates (Fig. 3C). Interestingly, presence of genes involved in siderophore biosynthesis was also identified in the bacterial genome assemblage to substantiate the siderophore detection assay (Fig. 3D). It is of interest to study the microbial ecology with respect of inter specific competition between the screened isolated microbes, and might be attempted in the future studies, such as metagenomic analysis.

Reactive oxygen species (ROS) are developed as intermediates of metal-catalyzed reaction by-products in the course of metabolic reactions encompassing mitochondrial electron transport, wherein ferrous ion (Fe^{2+}) holds the capacity to propagate the free radical chain reaction by acquiring or losing electrons. Consequently, the diminution of the development of ROS could be accomplished by Fe^{2+} ion chelation potential of *B. velezensis* MTCC 13048 (IC_{50} 4.2 μg/mL), which could contribute to antioxidant property for iron-related pathways. Hamidi et al. (2019) reported that the marine heterotrophic bacteria, such as *B. amyloliquefacines* could produce exopolysaccharides exhibiting 64% Fe^{2+} chelation with a concentration of 100 μg/mL. Likewise, the presently studied bacterium displayed potentially greater Fe^{2+} chelating properties. Coalescing the observations from the study, the production of siderophore metabolites and siderophore biosynthetic genes for bacillibactin (MT682353) and enterobactin (MT682354) in conjunction with the moderate Fe^{2+} chelation recognized its potential to scavenge iron from the surrounding environment, and thus creating a competition to the pathogenic bacteria along with reducing the ROS accumulation in the body. Additionally, siderophore conjugated antibiotics were reported to be easily transported into the cells of Gram-negative bacteria by exploiting their iron acquisition pathways. The biocontrol mechanism is dependent on the competition between transferrin molecules by pathogens and siderophores produced by bioactive microorganism in forming complexes with iron, and the siderophore have the advantage on account of its higher iron stability constants (Gram et al. 1999).

The inhibition curve was parallel to the growth curve, which could recognize that the inhibition per number of cells in exponential growth appeared to be stable throughout the growth, and it could be inferred that the more cells we have, the more secondary metabolites are produced. Ethyl acetate-derived organic extract of *B. velezensis* MBTDLP1 (MTCC13048) exhibited significant antibacterial activities against the test pathogens, and inhibition zone was ranged between 13 and 25 mm, even against the drug-resistant strain, MRSA, whereas chloramphenicol showed comparatively lesser zone of inhibition against these pathogens (Fig. 4B-D). The MIC of the organic extract was found to be 7.5 μg/mL against test pathogens, such as *V. parahemolyticus* and *S. pyogenes* (Table 3), and chloramphenicol showed a comparable MIC of 6.25 μg/mL against MRSA, *V. parahemolyticus* and 12.5 μg/mL against *S. pyogenes.* In the present study, disc diffusion assay showed lesser zone of inhibition when compared with spot-on-lawn analysis. This could be explained by the differential ability of the extract in diffusing from discs to the agar media, as the compounds might be sturdily adhered to cellulose discs. The ordinary filter paper utilized in disc diffusion experiment is made of cellulosic material [i.e., $\beta(1\rightarrow4)$ connected glucose units], wherein the free –OH groups on the polysaccharide could impart hydrophilicity. In consequence, if the compounds present in the organic extract of *B. velezensis* were of high polar in nature, they were anticipated to adsorb to the disc, and not diffused into the agar medium. Resultantly, a strain could possibly produce bioactive compounds, which are highly polar in character, but could not proportionally display antibacterial characteristics by disc diffusion technique, owing to the reasons put forth earlier (Burgess et al. 1999).

Organic ethyl acetate extract of the bacterium was used to perform NMR spectroscopic experiments, to accentuate the thumbprint regions, and to elucidate the preliminary structural information of the compounds. Apparently, greater number of proton integral at the deshielded section of the ^{1}H-NMR spectrum could infer the presence of polar functional groups, which might occupy predominant roles to exhibit prospective antimicrobial activities. ^{1}H-NMR integral of the organic extract was higher at δ_H 3.6–4.5 (proton integral $\sum H$ 0.85) (Fig. S4), which steered us to infer that the bioactive compounds in the extract might be highly polar in nature. Additionally, the protons at characterized regions of the ^{1}H-NMR spectrum of the purified fraction BV-1 isolated from the bacterial extract (Table S2) were combined to acquire the aggregate number of protons in the corresponding sections. Total number of protons at δ_H 4.6–6.5, owing to the presence of olefinic/alkanoates, were significantly greater ($\sum H$ 0.74) in the purified BV-1, whereas the crude extract displayed significantly lesser proton integral value ($\sum H$ 0.22), which concluded that the purification process of the crude could isolate the compound with greater electronegative olefinic/alkanoate functional groups (Table S2). Apparently, greater proton integral at the deshielded expanse of the ^{1}H-NMR spectrum in the purified compound (BV-1) could infer the presence of polar functional groups, which might occupy predominant roles to exhibit prospective antimicrobial properties.

Structure of amido-type 12-membered macrocyclic polyketide, 8-(pent-2-enyl)-1-oxo-5a, 8a-dioxacyclododecanyl-3-oxy-ethyl-5b′-methyl-5′-(7′(9′(methoxyethyl)-dihydrofuranyl) propanmido) succinate was deduced as $C_{32}H_{47}NO_{12}$ {HRESIMS: *m/z* 638.3181 $[M+H]^+$} (Fig. S5, Fig. 4E–F). The ^{13}C-NMR peak recorded in the deshielded region δ_C 209.16 and 173.87 was corroborated to the keto and amide carbonyl carbons, respectively, whereas other downfield signals at δ_C 170.08, 171.28, 167.26 could correspond to the ester carbonyl functionality. Hetereonuclear HMBC cross peaks between δ_H 3.39 (H-8) with δ_C 142.49 (H-8^1) supported the attachment to the C-8 location of 12-membered ring in the titled compound. Significantly deshielded proton doublet signal at δ_H 8.01 (H-6′) corresponds to an amide proton, which was further confirmed by HMBC correlation of doublet proton at δ_H 8.01 (H-6′) to the adjacent carbonyl carbon δ_C 173.87 (C-7′) and methine carbon δ_C 56.60 (C-5′). The connection of pentenyl group at C-8 position was confirmed through the HMBC correlations of δ_H 3.39 (H-8) with δ_C 142.49 (C-8^1). Correspondingly, the oxo-linked attachment of long chain containing amide linkage to the C-3 position of macrocylic ring corroborated with the HMBC cross-peak at δ_H 3.65 (H-1′) with δ_C 70.40 (C-3) in the downfield region. The heteronuclear correlations from δ_H 5.01 (H-13′) to δ_C 49.36 (C-12′), δ_C 167.26 (C-11′) and δ_C 149.90 (C-10'); δ_H 2.30 (H-9′) to δ_C 149.90 (C-10') together with one hydrogen correlations between the protons H-12′ (δ_H 2.96) and H-13′ (δ_H 5.01) attributed the dihydrofuranyl ring. Larger coupling constants of olefinic protons at H-4 (δ_H 4.90, $J=12.5$ Hz) and H-5 (δ_H 6.24, $J=13.2$ Hz) inferred that these protons were aligned in *cis* orientation, and were assigned as *Z*.

Notably, the MIC of the organic extract (7.5 µg/mL) was found to be analogous with that of the standard antibiotics (6.25 µg/mL), against most of the pathogens. The micro-dilution method displayed potential MIC with the purified polyketide against methicillin-resistant *S. aureus* (ATCC 33592) (MIC 0.38 µg/mL) (Table 2), than that exhibited by positive control chloramphenicol (6.25 µg/mL). A polyketide-originated macrolactin characterized as 8-(pent-2-enyl)-1-oxo-5a,8a-dioxacyclododecanyl-3-oxy-ethyl-5b′-methyl-5′-(7′(9′ (methoxyethyl)-dihydrofuranyl) propanmido) succinate was isolated from the crude extract. The purified compound displayed potential antibacterial potential against methicillin-resistant *S. aureus* (MIC 0.38 µg/mL). Forthcoming studies will envisage detailed bioactivity assessment and scaling up of the compound from *B. velezensis* MBTDLP1 (MTCC 13048) in the direction to develop potential antimicrobial agent.

Conclusion

In the search for novel antibiotics from marine macroalgae-associated symbionts, heterotrophic *B. velezensis* MBTDLP1 (MTCC accession number 13048) isolated from marine macroalga showed promising antimicrobial and iron-chelating activities with the supportive genetic backup of *pks* specific genes, siderophore biosynthetic genes, and could be exploited further for their advanced pharmacological evaluation. Chromatographic separation of organic extract of *B. velezensis* afforded 8-(pent-2-enyl)-1-oxo-5a,8a-dioxacyclododecanyl-3-oxy-ethyl-5b′-methyl-5′-(7′(9′ (methoxyethyl)-dihydrofuranyl)propanmido) succinate with antibacterial potential against methicillin-resistant *S. aureus*. From these results, the purified polyketide-originated macrolactin be worthy of growing consideration as a source of potential antibacterial agent for biotechnological applications.

GenBank accession numbers

Whole genome sequence of *B. velezensis* (MBTDLP1, MTCC 13048) was deposited in the NCBI database (accession number JABZEG000000000). Partial 16S rRNA gene sequences of the two bacterial isolates were submited in NCBI GenBank (accession numbers of MT122835, MT122905), and the partial *pks* sequences corresponding to *B. velezensis* MBTDLP1 was deposited with the accession number MT394492. Coding sequences for siderophore biosynthetic genes retrieved from whole genome data were deposited in Genbank with the accession numbers MT682354 and MT682353.

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Authors contributions KC and AF conceived and designed research, acquired funds, and conducted experiments. KC, RDC, SA, and VKK analyzed data. KC, RDC, SKP, and VKK drafted the manuscript. All authors read and approved the manuscript.

Declarations

Conflict of interest No potential conflict of interest was reported by the authors. This article does not contain any studies with human participants or animals performed by any of the authors.

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